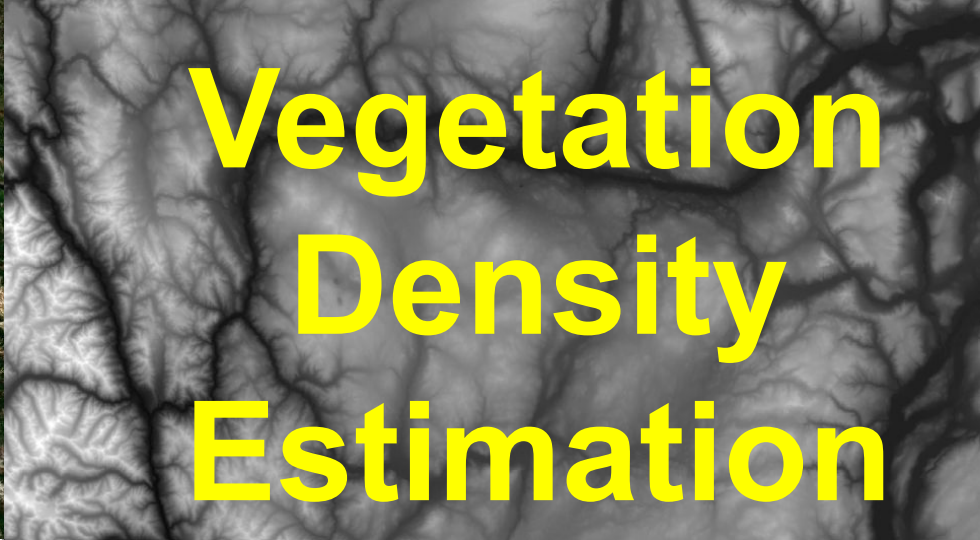
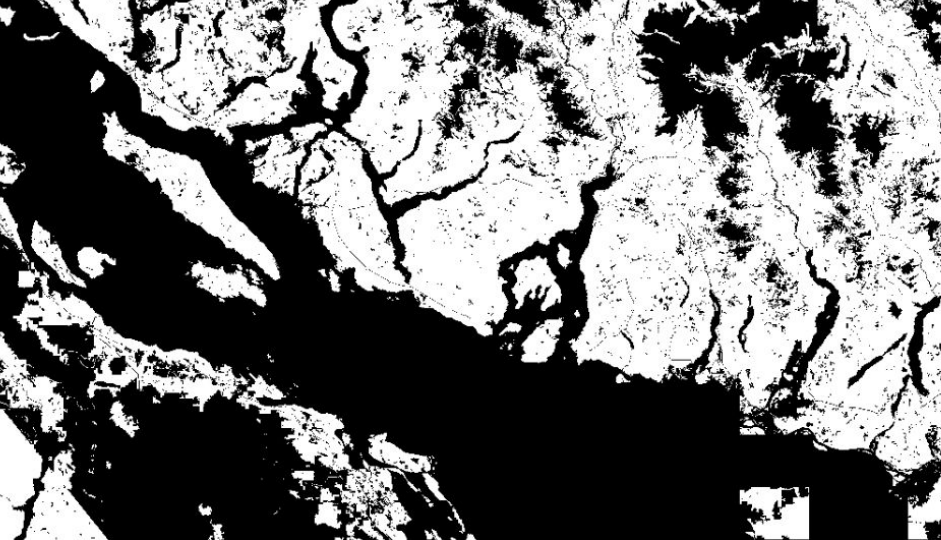




Vegetation Density Estimation



Jamie Breault

Understanding Vegetation Density Estimation

Prediction criteria? Regression or classification?

- Crown closure, regression

Target data?

- VRI - Forest Vegetation Composite R1 (crown_closure)

Questions for a forestry professional?

- Predictability vs interpretability? Goal metrics?

Key assumptions?

- The VRI dataset is assumed to be accurate
- The area used for training is assumed to be representative

Area of Interest (AOI)

- 95,580 km²
 - Latitude: 49°N to 51°N
 - Longitude: 119°W to 125°W
- Representative of much of BC
 - Rainforests
 - Arid forests
 - Mountains
 - Ocean, lakes, rivers
 - Cities
 - Farms



Data Sources

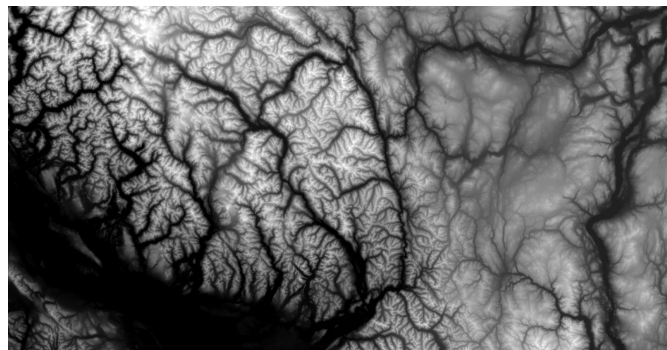
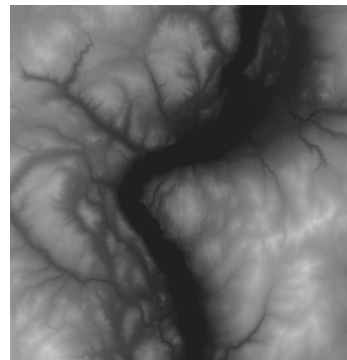
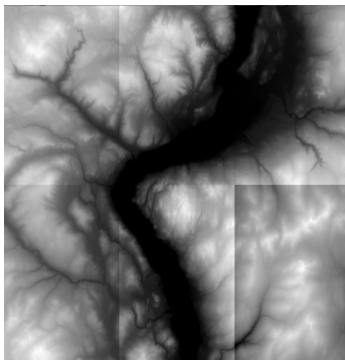
	Sentinel-2 Cloudless	BC CDED (DEM)	VRI - Forest Vegetation Composite R1
Description	Satellite imagery (RGB)	Elevation	Crown closure (and more)
Resolution	10 m/pixel	0.75 arc-sec (~23 m)	Vector polygons
CRS	EPSG:3857	EPSG:4269	EPSG:3005
Year	2025	2012	2025
Range	Global	BC	BC

Elevation

1. Web scrape for DEM patches that cover AOI
 - Each folder contains 8-32 DEM patches
2. Merge DEM patches into one layer
3. Crop to AOI

Index of /datasets/175624

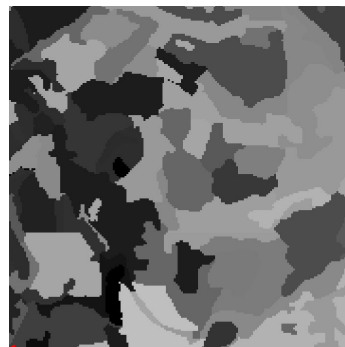
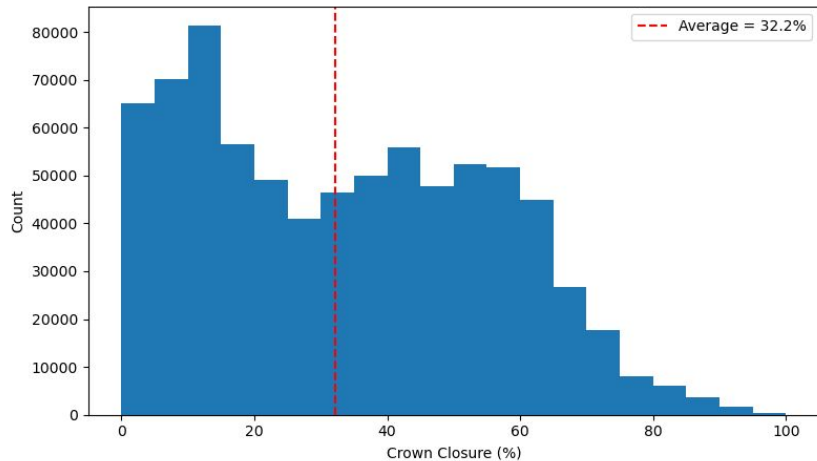
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94m/	2018-04-12 12:41
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94k/	2018-04-12 12:41
94j/	2018-04-12 12:41
94i/	2018-04-12 12:41
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94a/	2018-04-12 12:40
93p/	2018-04-12 12:40
93o/	2018-04-12 12:40
93n/	2018-04-12 12:40



Crown Closure

1. Download VRI dataset
2. Crop polygons to AOI
3. Make raster of crown closure
 - Null values are treated as 0

Crown Closure Distribution



Patches

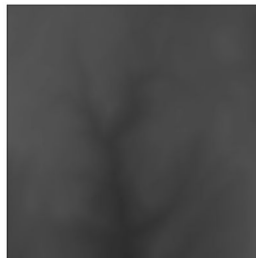
- Patches are arranged in a grid filling the AOI
- For every patch, there is an image for:
 - Satellite imagery
 - Elevation
 - Crown closure
- Elevation is normalized against the highest point in BC (4,671 m)

Pixel Size	25 m
Patch Dimensions	6.4 km × 6.4 km
Number of Patches	5616

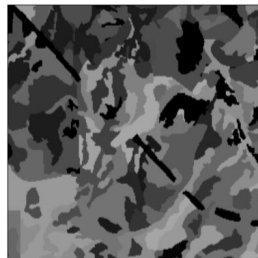
Satellite



Elevation



Crown Closure



Linear Regression

1. For each patch, save a CSV of each pixel's RGB, elevation, and crown closure
 - 368 million pixels, 20.4 GB
2. Perform linear regression one patch at a time
 - 80% of each patch for training, 20% for testing

$$\text{crown_closure} = 12 - 154(R) + 96(G) + 28(B) + 144(\text{elevation}) - 266(\text{elevation}^2)$$

Test RMSE	20.3 %
Inference Time Per Image	0.39 ms

SegFormer

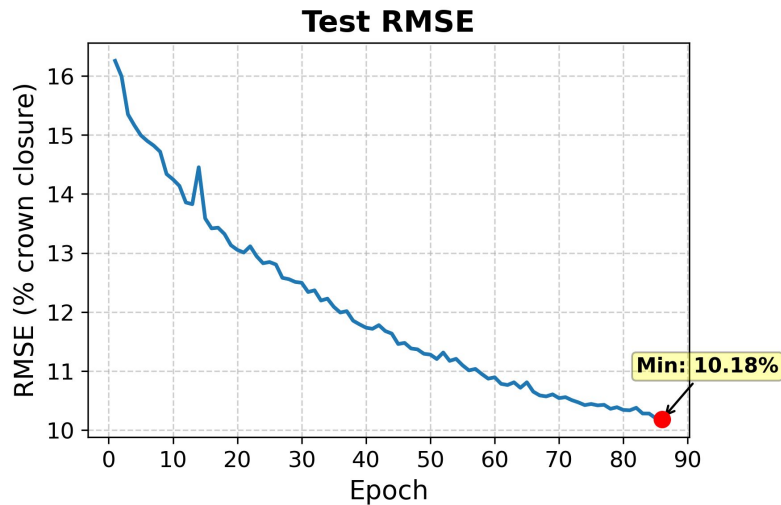
1. Set up pretrained model

- **Name:** PranilIIIII/segformer-satellite-segementation
- Pretrained on satellite imagery (RBG)
- First layer expanded from 3 to 4 input channels
- Classifier replaced by single-channel regression head

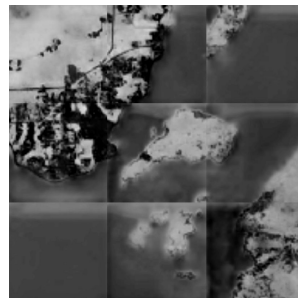
2. Train 86 epochs

- 80% of each patch for training, 20% for testing

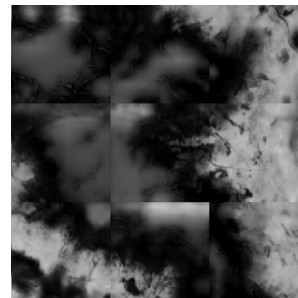
	Linear Regression	SegFormer
Test RMSE	20.3 %	10.2 %
Inference Time Per Image	0.39 ms	6.26 ms



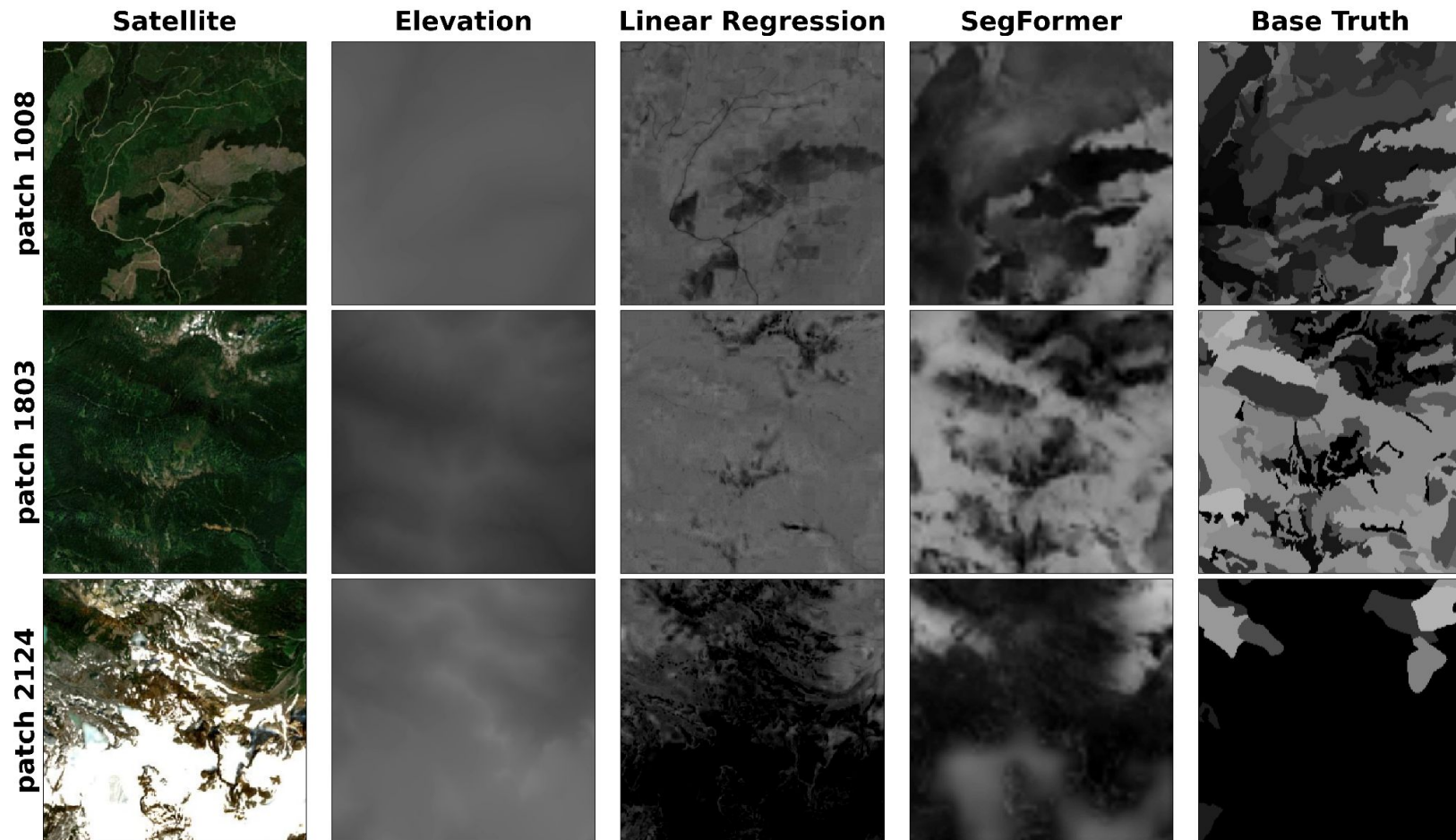
Coastline



Mountain Top



Comparing Crown Closure Estimations: Linear Regression vs. SegFormer



Next Steps

- Use validation set
- Separate training/validation/test sets by location
- Normalize elevation based on training data
- Overlap tiles
- Estimate null areas
- Use near-infrared
- More training
- More data

Null crown closure is black

